Quero que você me ajude a fazer o seguinte: eu quero desenvolver um projeto que, basicamente, vai ser um sisteminha bem simples, onde a pessoa insere o extrato bancário do mês que passou. O sistema analisa e subdivide em categorias: - investimentos - diversão - custos fixos - custos variáveis A partir disso, ele aplica alguma regra de investimento, como é falado por algum autor extremamente famoso, inglês, que seja um dos maiores investidores da história. A partir disso, ele faz uma análise de seu extrato, classifica quanto você gastou em cada parte e, a partir disso, mostra se, diminuindo isso, você consegue economizar mais e dá algumas sugestões de como você consegue aplicar mais em algum investimento seu. Seria isso ou ver padrões: você gasta tanto em gasolina, a gasolina não tem como ser de mercado, você gasta muito em conveniência, mercado, tal, tudo. Se você comprar tudo no atacado, você economiza mais. Essas coisinhas assim, algumas sugestões bem profundas, mas eu preciso que você me ajude a fazer esse projetinho. No caso, tem que ser uma tela simples, com uma forma de ele voltar e vir, seja bem intuitivo para qualquer leigo conseguir fazer e que só envie o extrato bancário. A partir do extrato bancário, ele aceita somente em PDF o documento, e a partir disso, ele faz uma análise, um resumo, e vai analisando a questão dos investimentos que você gasta todo mês, essas coisas.

Implemente a função classifyTransaction(description: string): Category seguindo estes exemplos de entrada/saída, baseados em formatos reais de extrato bancário brasileiro (note que cada banco formata de um jeito diferente — "PAG*", "COMPRA", "DEB AUT" — a função precisa lidar com essa variedade):

=== CUSTOS VARIÁVEIS (mercado, posto, farmácia — valor oscila a cada compra) ===
Entrada: "PAG*SUPERMUFFATO 4521" → Saída: "Custos Variáveis"
Entrada: "COMPRA CONDOR SUPERMERCADOS" → Saída: "Custos Variáveis"
Entrada: "PAG*CIDADE CANCAO SUPERM" → Saída: "Custos Variáveis"
Entrada: "COMPRA BOA COMPRA SUPERM" → Saída: "Custos Variáveis"
Entrada: "PAG*IGUACU SUPERMERCADO" → Saída: "Custos Variáveis"
Entrada: "COMPRA CARREFOUR BR" → Saída: "Custos Variáveis"
Entrada: "PAG*EXTRA HIPERMERCADO" → Saída: "Custos Variáveis"
Entrada: "COMPRA PAO DE ACUCAR" → Saída: "Custos Variáveis"
Entrada: "PAG*ASSAI ATACADISTA" → Saída: "Custos Variáveis"
Entrada: "COMPRA ATACADAO LTDA" → Saída: "Custos Variáveis"
Entrada: "PAG*DIA SUPERMERCADO" → Saída: "Custos Variáveis"
Entrada: "COMPRA SUPERMERCADO BIG" → Saída: "Custos Variáveis"
Entrada: "PAG*POSTO IPIRANGA MGA" → Saída: "Custos Variáveis"
Entrada: "COMPRA POSTO SHELL SELECT" → Saída: "Custos Variáveis"
Entrada: "PAG*POSTO BR MANDACARU" → Saída: "Custos Variáveis"
Entrada: "COMPRA AUTO POSTO ALE" → Saída: "Custos Variáveis"
Entrada: "PAG*POSTO RAIZEN" → Saída: "Custos Variáveis"
Entrada: "COMPRA DROGARIA SAO PAULO" → Saída: "Custos Variáveis"
Entrada: "PAG*DROGASIL 0231" → Saída: "Custos Variáveis"

Entrada: "COMPRA FARMACIA PACHECO" → Saída: "Custos Variáveis"
Entrada: "PAG*ULTRAFARMA" → Saída: "Custos Variáveis"
Entrada: "COMPRA PAGUE MENOS FARM" → Saída: "Custos Variáveis"
Entrada: "PAG*PADARIA DOCE MEL" → Saída: "Custos Variáveis"
Entrada: "COMPRA PADARIA SAO JOSE" → Saída: "Custos Variáveis"
Entrada: "PAG*ACOUGUE BOM JESUS" → Saída: "Custos Variáveis"
Entrada: "COMPRA ACOUGUE NOBRE CARNE" → Saída: "Custos Variáveis"
Entrada: "PAG*HORTIFRUTI RAIZ" → Saída: "Custos Variáveis"
Entrada: "COMPRA SACOLAO CENTRAL" → Saída: "Custos Variáveis"
Entrada: "DEB AUT SANEPAR AGUA" → Saída: "Custos Variáveis"
Entrada: "DEB AUT COPEL ENERGIA" → Saída: "Custos Variáveis"
Entrada: "DEB AUT COMPANHIA GAS PR" → Saída: "Custos Variáveis"
Entrada: "PAG*QUITANDA DO BAIRRO" → Saída: "Custos Variáveis"
Entrada: "COMPRA FEIRA LIVRE" → Saída: "Custos Variáveis"
Entrada: "PAG*PANIFICADORA STA CLARA" → Saída: "Custos Variáveis"
Entrada: "COMPRA MERCADINHO IRMAOS" → Saída: "Custos Variáveis"
Entrada: "PAG*POSTO GRAAL RODOVIA" → Saída: "Custos Variáveis"
Entrada: "COMPRA LOJA CONVENIENCIA BR" → Saída: "Custos Variáveis"
Entrada: "PAG*OXXO LOJA 234" → Saída: "Custos Variáveis"
Entrada: "COMPRA AM/PM CONVENIENCIA" → Saída: "Custos Variáveis"
Entrada: "PAG*MERCEARIA DO ZE" → Saída: "Custos Variáveis"
Entrada: "COMPRA VAREJAO CENTRAL" → Saída: "Custos Variáveis"
Entrada: "PAG*SUPER NOSSO" → Saída: "Custos Variáveis"
Entrada: "COMPRA COMPER SUPERMERCADOS" → Saída: "Custos Variáveis"
Entrada: "PAG*BRETAS SUPERMERCADOS" → Saída: "Custos Variáveis"
Entrada: "COMPRA SAVEGNAGO SUPERMERCADOS" → Saída: "Custos Variáveis"
Entrada: "PAG*ANGELONI SUPERMERCADOS" → Saída: "Custos Variáveis"
Entrada: "COMPRA MUNDIAL SUPERMERCADOS" → Saída: "Custos Variáveis"
Entrada: "PAG*FARMACIA SAO JOAO" → Saída: "Custos Variáveis"
Entrada: "COMPRA DROGA RAIA" → Saída: "Custos Variáveis"
Entrada: "PAG*POSTO EQUADOR" → Saída: "Custos Variáveis"

=== CUSTOS FIXOS (assinaturas, contas recorrentes — valor previsível todo mês) ===
Entrada: "NETFLIX.COM" → Saída: "Custos Fixos"
Entrada: "SPOTIFY AB" → Saída: "Custos Fixos"
Entrada: "DISNEY PLUS" → Saída: "Custos Fixos"
Entrada: "AMAZON PRIME BR" → Saída: "Custos Fixos"
Entrada: "HBO MAX ASSINATURA" → Saída: "Custos Fixos"
Entrada: "YOUTUBE PREMIUM" → Saída: "Custos Fixos"
Entrada: "GLOBOPLAY ASSINATURA" → Saída: "Custos Fixos"
Entrada: "DEEZER MUSIC" → Saída: "Custos Fixos"
Entrada: "APPLE.COM/BILL" → Saída: "Custos Fixos"
Entrada: "MENSALIDADE UNICESUMAR" → Saída: "Custos Fixos"
Entrada: "MENSALIDADE ESTACIO EAD" → Saída: "Custos Fixos"

Entrada: "TAXA CONDOMINIO ED RESIDENCIAL" → Saída: "Custos Fixos"
Entrada: "ALUGUEL IMOVEL RESIDENCIAL" → Saída: "Custos Fixos"
Entrada: "FINANCIAMENTO IMOBILIARIO CAIXA" → Saída: "Custos Fixos"
Entrada: "FINANCIAMENTO VEICULO SANTANDER" → Saída: "Custos Fixos"
Entrada: "CONSORCIO CONTEMPLADO" → Saída: "Custos Fixos"
Entrada: "SEGURO AUTO PORTO SEGURO" → Saída: "Custos Fixos"
Entrada: "SEGURO RESIDENCIAL BRADESCO" → Saída: "Custos Fixos"
Entrada: "SEGURO VIDA SULAMERICA" → Saída: "Custos Fixos"
Entrada: "PLANO SAUDE UNIMED MARINGA" → Saída: "Custos Fixos"
Entrada: "PLANO SAUDE AMIL" → Saída: "Custos Fixos"
Entrada: "PLANO SAUDE HAPVIDA" → Saída: "Custos Fixos"
Entrada: "ACADEMIA SMARTFIT MARINGA" → Saída: "Custos Fixos"
Entrada: "ACADEMIA BIO RITMO" → Saída: "Custos Fixos"
Entrada: "ACADEMIA BODYTECH" → Saída: "Custos Fixos"
Entrada: "VIVO TELEFONIA MOVEL" → Saída: "Custos Fixos"
Entrada: "CLARO NET TELEFONIA" → Saída: "Custos Fixos"
Entrada: "TIM CELULAR MENSALIDADE" → Saída: "Custos Fixos"
Entrada: "OI FIBRA INTERNET" → Saída: "Custos Fixos"
Entrada: "DEB AUT MENSALIDADE ESCOLA" → Saída: "Custos Fixos"
Entrada: "IPTU PARCELADO" → Saída: "Custos Fixos"
Entrada: "IPVA PARCELADO" → Saída: "Custos Fixos"
Entrada: "CARTAO CONSIGNADO INSS" → Saída: "Custos Fixos"
Entrada: "EMPRESTIMO PESSOAL PARCELA" → Saída: "Custos Fixos"
Entrada: "MICROSOFT 365 ASSINATURA" → Saída: "Custos Fixos"
Entrada: "ADOBE CREATIVE CLOUD" → Saída: "Custos Fixos"
Entrada: "ICLOUD STORAGE" → Saída: "Custos Fixos"
Entrada: "XBOX GAME PASS" → Saída: "Custos Fixos"
Entrada: "PLAYSTATION PLUS" → Saída: "Custos Fixos"
Entrada: "ESTACIONAMENTO MENSALISTA" → Saída: "Custos Fixos"
Entrada: "PLANO ODONTOLOGICO ODONTOPREV" → Saída: "Custos Fixos"
Entrada: "CLUBE ASSINATURA BARBEARIA" → Saída: "Custos Fixos"
Entrada: "MENSALIDADE INGLES WIZARD" → Saída: "Custos Fixos"
Entrada: "MENSALIDADE CURSO IDIOMAS CCAA" → Saída: "Custos Fixos"
Entrada: "DEB AUT SEGURO CELULAR" → Saída: "Custos Fixos"
Entrada: "TAXA MANUTENCAO CONTA" → Saída: "Custos Fixos"
Entrada: "ANUIDADE CARTAO CREDITO" → Saída: "Custos Fixos"
Entrada: "MENSALIDADE PLANO CELULAR CONTROLE" → Saída: "Custos Fixos"
Entrada: "ALARME MONITORADO VERISURE" → Saída: "Custos Fixos"
Entrada: "TV POR ASSINATURA SKY" → Saída: "Custos Fixos"

=== INVESTIMENTOS (aportes, corretoras, produtos financeiros) ===
Entrada: "XP INVESTIMENTOS CORRETORA" → Saída: "Investimentos"
Entrada: "RICO INVESTIMENTOS" → Saída: "Investimentos"
Entrada: "CLEAR CORRETORA" → Saída: "Investimentos"

Entrada: "NUINVEST TRANSFERENCIA" → Saída: "Investimentos"
Entrada: "BTG PACTUAL DIGITAL" → Saída: "Investimentos"
Entrada: "INTER INVEST APLICACAO" → Saída: "Investimentos"
Entrada: "MODALMAIS CORRETORA" → Saída: "Investimentos"
Entrada: "TORO INVESTIMENTOS" → Saída: "Investimentos"
Entrada: "AGORA INVESTIMENTOS" → Saída: "Investimentos"
Entrada: "GENIAL INVESTIMENTOS" → Saída: "Investimentos"
Entrada: "TESOURO DIRETO APLICACAO" → Saída: "Investimentos"
Entrada: "TESOURO SELIC COMPRA" → Saída: "Investimentos"
Entrada: "CDB BANCO INTER" → Saída: "Investimentos"
Entrada: "CDB NUBANK APLICACAO" → Saída: "Investimentos"
Entrada: "LCI CAIXA ECONOMICA" → Saída: "Investimentos"
Entrada: "LCA BANCO DO BRASIL" → Saída: "Investimentos"
Entrada: "PREVIDENCIA PRIVADA BRADESCO" → Saída: "Investimentos"
Entrada: "PREVIDENCIA ITAU PRIVATE" → Saída: "Investimentos"
Entrada: "PREVIDENCIA PORTO SEGURO" → Saída: "Investimentos"
Entrada: "FUNDO IMOBILIARIO XP" → Saída: "Investimentos"
Entrada: "FUNDO DE INVESTIMENTO ITAU" → Saída: "Investimentos"
Entrada: "APORTE FUNDO RENDA FIXA" → Saída: "Investimentos"
Entrada: "COMPRA ACOES B3" → Saída: "Investimentos"
Entrada: "TRANSFERENCIA CORRETORA VALOR" → Saída: "Investimentos"
Entrada: "APLICACAO POUPANCA PROGRAMADA" → Saída: "Investimentos"
Entrada: "PICPAY INVEST APLICACAO" → Saída: "Investimentos"
Entrada: "MERCADO BITCOIN COMPRA" → Saída: "Investimentos"
Entrada: "BINANCE COMPRA CRIPTO" → Saída: "Investimentos"
Entrada: "FOXBIT CORRETORA CRIPTO" → Saída: "Investimentos"
Entrada: "NOVADAX INVESTIMENTOS" → Saída: "Investimentos"
Entrada: "WARREN INVESTIMENTOS" → Saída: "Investimentos"
Entrada: "VITREO INVESTIMENTOS" → Saída: "Investimentos"
Entrada: "EASYNVEST APLICACAO" → Saída: "Investimentos"
Entrada: "ORAMA INVESTIMENTOS" → Saída: "Investimentos"
Entrada: "SAFRA INVEST APLICACAO" → Saída: "Investimentos"
Entrada: "C6 INVEST APLICACAO" → Saída: "Investimentos"
Entrada: "BANCO INTER TESOURO" → Saída: "Investimentos"
Entrada: "NU INVEST RENDA FIXA" → Saída: "Investimentos"
Entrada: "XP PREVIDENCIA APLICACAO" → Saída: "Investimentos"
Entrada: "APLICACAO FUNDO DI" → Saída: "Investimentos"
Entrada: "APLICACAO FUNDO MULTIMERCADO" → Saída: "Investimentos"
Entrada: "COMPRA COTAS FII" → Saída: "Investimentos"
Entrada: "DEBENTURE INCENTIVADA COMPRA" → Saída: "Investimentos"
Entrada: "CRA APLICACAO" → Saída: "Investimentos"
Entrada: "CRI APLICACAO" → Saída: "Investimentos"
Entrada: "POUPANCA PROGRAMADA CAIXA" → Saída: "Investimentos"
Entrada: "TRANSFERENCIA TED CORRETORA" → Saída: "Investimentos"

Entrada: "APORTE MENSAL PREVIDENCIA" → Saída: "Investimentos"
Entrada: "COMPRA ETF BOVA11" → Saída: "Investimentos"
Entrada: "APLICACAO RENDA FIXA SOFISTICADA" → Saída: "Investimentos"

=== OUTROS (lazer, compras não essenciais, ou não reconhecido) ===
Entrada: "IFOOD *PEDIDO 88213" → Saída: "Outros"
Entrada: "UBER *TRIP" → Saída: "Outros"
Entrada: "UBER EATS PEDIDO" → Saída: "Outros"
Entrada: "99POP CORRIDA" → Saída: "Outros"
Entrada: "RAPPI PEDIDO" → Saída: "Outros"
Entrada: "PAG*SHOPPING CIDADE CANCAO" → Saída: "Outros"
Entrada: "PAG*SHOPPING AVENIDA CENTER" → Saída: "Outros"
Entrada: "PAG*HAVAN MARINGA" → Saída: "Outros"
Entrada: "CINEMARK SHOPPING AVENIDA" → Saída: "Outros"
Entrada: "UCI CINEMAS" → Saída: "Outros"
Entrada: "BAR DO ZE MARINGA" → Saída: "Outros"
Entrada: "RESTAURANTE OUTBACK" → Saída: "Outros"
Entrada: "CHOPERIA CENTRAL" → Saída: "Outros"
Entrada: "COMPRA MERCADO LIVRE" → Saída: "Outros"
Entrada: "COMPRA AMAZON.COM.BR" → Saída: "Outros"
Entrada: "COMPRA SHEIN" → Saída: "Outros"
Entrada: "COMPRA SHOPEE" → Saída: "Outros"
Entrada: "COMPRA ALIEXPRESS" → Saída: "Outros"
Entrada: "LOJA RENNER" → Saída: "Outros"
Entrada: "LOJA C&A" → Saída: "Outros"
Entrada: "LOJA RIACHUELO" → Saída: "Outros"
Entrada: "LOJA ZARA BRASIL" → Saída: "Outros"
Entrada: "LIVRARIA CULTURA" → Saída: "Outros"
Entrada: "LIVRARIA SARAIVA" → Saída: "Outros"
Entrada: "STEAM GAMES" → Saída: "Outros"
Entrada: "PLAYSTATION STORE COMPRA" → Saída: "Outros"
Entrada: "XBOX LIVE COMPRA" → Saída: "Outros"
Entrada: "INGRESSO.COM COMPRA" → Saída: "Outros"
Entrada: "SYMPLA EVENTOS" → Saída: "Outros"
Entrada: "123 MILHAS PACOTE" → Saída: "Outros"
Entrada: "DECOLAR.COM RESERVA" → Saída: "Outros"
Entrada: "LATAM VIAGENS" → Saída: "Outros"
Entrada: "AZUL LINHAS AEREAS" → Saída: "Outros"
Entrada: "AIRBNB RESERVA" → Saída: "Outros"
Entrada: "BOOKING.COM RESERVA" → Saída: "Outros"
Entrada: "BALADA NOTURNA CLUB" → Saída: "Outros"
Entrada: "CASA DE SHOW MARINGA" → Saída: "Outros"
Entrada: "SORVETERIA GELATO" → Saída: "Outros"
Entrada: "HAMBURGUERIA ARTESANAL" → Saída: "Outros"

Entrada: "PIZZARIA DELIVERY" → Saída: "Outros"
Entrada: "STARBUCKS CAFE" → Saída: "Outros"
Entrada: "CAFETERIA CENTRO" → Saída: "Outros"
Entrada: "LOJA DE PRESENTES" → Saída: "Outros"
Entrada: "FLORICULTURA ENTREGA" → Saída: "Outros"
Entrada: "PET SHOP RACAO PREMIUM" → Saída: "Outros"
Entrada: "SALAO DE BELEZA" → Saída: "Outros"
Entrada: "BARBEARIA AVULSA" → Saída: "Outros"
Entrada: "TATUAGEM ESTUDIO" → Saída: "Outros"
Entrada: "LOJA DE ELETRONICOS MAGAZINE" → Saída: "Outros"
Entrada: "COMPRA GENERICA NAO IDENTIFICADA" → Saída: "Outros"

Implemente a função completa em TypeScript, cobrindo esses padrões via dicionário de palavras-chave (case-insensitive, ignorando acentos e números/códigos soltos no fim da descrição) e generalizando pra casos parecidos que não estão na lista acima. Se nada bater, retorne "Outros". Depois de implementar, escreva também 10-15 testes automatizados cobrindo pelo menos 3 exemplos de cada categoria (incluindo alguns que não estavam na lista de exemplos, pra provar que a função generaliza) e casos de borda (descrição vazia, tudo maiúsculo, tudo minúsculo, com acentos).

I'm designing a cascading rule pipeline for classification, where priority order matters since similar keywords can map to different categories—like distinguishing "PLAYSTATION PLUS" (Custos Fixos) from "PLAYSTATION STORE COMPRA" (Outros), so I need precise keyword matching rather than simple substring checks.

I'm working through category priority: Investimentos, Custos Fixos, Custos Variáveis, then Outros as fallback, while watching for tricky overlaps—like "COMPRA" appearing in investment purchases (ACOES, FII, ETF) versus generic purchases, and especially the "MERCADO" problem where "SUPERMERCADO" should match Custos Variáveis but "MERCADO LIVRE" should match Outros, meaning I need careful token-based or normalized string matching rather than naive substring checks to avoid false positives.

Since "BITCOIN" is the distinguishing keyword, it'll match before any generic "MERCADO" term, so I don't need "MERCADO" as a standalone keyword at all. "MERCADO LIVRE" won't match "SUPERMERCADO" as a substring, so it correctly falls through to Outros by default.

Now I need to watch out for "COMPRA" showing up generically across Investimentos entries too, like purchases of stocks, fund shares, or ETFs. I'll rely on more specific markers like "ACOES", "B3", "FII", "ETF", or ticker-like strings such as "BOVA11" to distinguish these, though generalizing to arbitrary ETF tickers could get tricky.

Given this is meant to be a cascading rule-based classifier, I'll keep it simple with an ordered list of category rules where the first match wins, and I'm sketching out a normalization step that uppercases text and strips accents before matching.

Next I need to strip trailing numeric codes like store numbers, since the instructions specify ignoring accents and loose trailing digits in descriptions. Prefixes like "PAG*" or "COMPRA" can stay since keyword matching targets the core brand name, not the prefix.

For tickers like "BOVA11", stripping trailing digits would leave "BOVA", but since "ETF" already serves as the keyword match, I don't need ticker-specific codes at all—better to rely on generalized keywords. I'm also considering ambiguous 3-letter codes like "CRA" and "CRI" for investment applications.

To avoid false positives where short keywords accidentally match inside longer words, I'll use word-boundary matching by padding the normalized string with spaces and searching for the keyword wrapped in spaces, rather than plain substring matching—keeping things simple while still being accurate for short ambiguous terms.

Now I'm sketching the normalize function: strip accents, uppercase everything, and collapse non-alphanumeric characters into spaces so merchant strings like "PAG*SUPERMUFFATO" or "IFOOD *PEDIDO 88213" become clean space-separated tokens.

Since Outros is the natural fallback, stores like RENNER, RIACHUELO, and ZARA will correctly default there without needing explicit keywords, which saves me from building exhaustive lists. I still want to verify none of these Outros examples accidentally trigger keywords from other categories, so I'm scanning through cases like "IFOOD *PEDID" to check for conflicts.

I'm checking that "PAG*SHOPPING CIDADE CANCAO" won't falsely match the supermarket keyword since it lacks "SUPERM" as a substring, unlike the variable-cost example that includes "SUPERM" explicitly - confirming my keyword design correctly catches variants like "SUPERMUFFATO" through that substring match.

To avoid false positives, I'll use the specific multi-word keyword "SUPER NOSSO" rather than the generic "SUPER" substring. Now I'm working through the keyword mapping for each category.

I'm designing a helper function that pads the normalized string and keyword with spaces before checking for substring inclusion, so matches only occur on whole-word boundaries rather than inside other words.

Word-boundary matching won't catch "SUPERMUFFATO" unless explicitly listed, so I'm reconsidering: since the task wants generalization to similar unlisted chains, plain substring matching might actually work better than strict word-boundary matching, since many supermarket names share "SUPER" or "MERCADO" as prefixes or suffixes within compound names.

I'm deciding to use substring matching for keyword stems, but need to pick stems carefully to avoid collisions with other categories. I'm building a list of stems for the groceries/pharmacy/gas station category: SUPERM, HIPERMERCADO, ATACAD, MERCEARIA, MERCADINHO, VAREJAO, SACOLAO, HORTIFRUTI, and checking each against potential false matches.

I'll use stem matching for pharmacy-related terms too: "DROG" covers Drogaria, Drogasil, and Droga Raia variants, while "FARM" catches Farmacia, Ultrafarma, and similar names. I'm also considering adding "SUPER" as a stem for Super Nosso, checking first that it doesn't collide with other categories.

Moving to convenience stores, I'm normalizing "AM/PM" to "AM PM" by stripping the slash, then starting to work through utility bill keywords like sanitation companies.

I'm noticing that "DEB AUT" prefix alone doesn't determine fixed vs variable cost—it's the specific keyword after it that matters, since water/energy/gas bills vary monthly while things like school tuition or insurance stay fixed despite sharing the same prefix. So I'm building out keywords like "SANEPAR", "COPEL", "AGUA", "ENERGIA" for variable utility costs, being careful with ambiguous stems like "GAS" that could collide with other terms.

I'll continue adding more streaming and subscription keywords like Amazon Prime, HBO, YouTube Premium, Globoplay, Deezer, Apple subscriptions, Microsoft 365, Adobe, iCloud, Xbox Game Pass, PlayStation Plus, and Sky TV. I need to be careful distinguishing "PLAYSTATION PLUS" (a fixed subscription) from "PLAYSTATION STORE" (a one-off purchase) by matching the full multi-word phrase rather than just "PLAYSTATION".

Similarly, I'm making sure "XBOX GAME PASS" is matched as a full phrase so it doesn't accidentally catch "XBOX LIVE COMPRA" purchases, and confirming "STEAM GAMES" falls into the miscellaneous category without needing a fixed-cost keyword.

For education expenses, I'm using the generic stem "MENSALIDADE" since it broadly covers tuition entries like university, school, language courses, and even phone plans - a nice generalization that works across many variations. Now I'm moving to housing costs, checking keywords like "CONDOMINIO" and rent-related terms.

I'll also add those brand keywords since they clearly signal a healthcare plan even without "PLANO" present. For gym, "ACADEMIA" as a stem works well since it covers chains like SMARTFIT and BODYTECH. For telecom, I'm considering VIVO, CLARO, TIM, and OI as carrier keywords, checking that short tokens like "OI" won't collide with other words when matched as standalone terms.

I'm deciding to use two matching modes: substring matching for stems meant to generalize prefixes (like SUPERM, DROG, FARM, ATACAD), and whole-word matching for short ambiguous tokens (like OI, TIM, GAS, B3) to avoid false positives.

I need to be careful picking keyword tokens so they don't collide across categories. Starting with investment-related terms, I'm checking that short tokens like "INVEST" cover most brokerage names, but noting exceptions like "BTG PACTUAL DIGITAL" which needs its own explicit keyword since it lacks the word "INVEST".

I notice XP PREVIDENCIA lacks INVEST but has PREVIDENCIA and APLICACAO, and EASYNVEST is tricky since it doesn't actually contain the literal "INVEST" substring—checking letter by letter confirms there's no "I" before "NVEST". I'll need to handle EASYNVEST as a special case separately.

Going through the rest of the list, ORAMA, WARREN, VITREO, and NOVADAX all have "INVESTIMENTOS" so they match INVEST fine. FOXBIT and BINANCE both contain CRIPTO, MERCADO BITCOIN has BITCOIN, and PICPAY INVEST APLICACAO checks out too.

I'm noting that "INVEST" as a substring catches most variants but misses EASYNVEST, so I'll add that as its own explicit keyword, along with BTG PACTUAL. For the remaining investment types, I'm planning stems like APLICACAO, TESOURO, CDB, LCI, LCA, and PREVIDENCIA to cover the various product names.

PREVIDENCIA privada contributions consistently count as investments rather than fixed costs across the test cases, even recurring ones like "APORTE MENSAL PREVIDENCIA," since they build wealth rather than being a pure expense. I'm also adding FUNDO as a stem to catch fund-related entries like "FUNDO IMOBILIARIO" and "FUNDO DE INVESTIMENTO."

I'll use "POUPANCA PROGRAMADA" as a multi-word stem to avoid misclassifying regular savings deposits, and add stems for fund/investment types like "FII", "ETF", "DEBENTURE", and carefully handle short tokens like "CRA"/"CRI" with word-boundary checks, plus crypto-related terms like "BITCOIN" and exchange names.

I'm confirming these stems are redundant with broader catches like INVESTIMENTOS/CORRETORA/INVEST/PREVIDENCIA, and that "TESOURO" covers various Tesouro Direto variants, then cross-checking that none of these investment keywords accidentally overlap with items in the Fixed/Variable/Other expense categories.

I'll skip the word-boundary complexity and just use plain substring matching since the dataset shows no collisions. I do need to be careful that my trailing-number-stripping regex doesn't accidentally remove something like "B3" since it's alphanumeric, not purely numeric, so the strip should only target pure digit groups.

Testing "PAG*SUPERMUFFATO 4521" confirms the trailing digit-stripping works correctly, and since keywords match as substrings anywhere, prefixes like "PAG*" don't need special handling. Now I'm planning the normalization pipeline: uppercase conversion, then diacritic removal via NFD normalization and stripping combining marks.

Next I'm replacing non-alphanumeric characters with spaces so symbols like *, ., /, &, - get converted, then collapsing multiple spaces and trimming, followed by stripping any trailing purely numeric token.

Now I'm designing the classifyTransaction function as a cascade: defining a Category type with options like Investimentos, Custos Fixos, Custos Variáveis, and Outros, building keyword lists for each category, and writing a helper that checks if normalized text includes any keyword from a given list.

The function normalizes the description, returns "Outros" if empty, then checks keyword matches in priority order—Investimentos first, then Custos Fixos, then Custos Variáveis, defaulting to Outros. I need to verify this priority order doesn't cause misclassification where keywords overlap between categories.

I'm tracing through edge cases: water bill entries correctly fall through to Variáveis since they don't match any Fixos keywords, condominium fees correctly match Fixos via "CONDOMINIO" without accidentally triggering Variáveis terms, and I'm checking whether "TAXA MANUTENCAO CONTA" needs an explicit "MANUTENCAO" keyword added to the Fixos list.

I'm deciding to add "CONDOMINIO" and "MANUTENCAO" as separate specific stems rather than a broad generic "TAXA" stem, to avoid false positives elsewhere, and confirming "ANUIDADE CARTAO CREDITO" correctly matches Fixos via the "ANUIDADE" keyword.

For "ALARME MONITORADO VERISURE" I'll use "ALARME" as the matching stem. For "TV POR ASSINATURA SKY" I'm checking that "ASSINATURA" works consistently across multiple Fixos entries like HBO MAX, GLOBOPLAY, MICROSOFT 365, and CLUBE ASSINATURA BARBEARIA, so generalizing with that stem makes sense.

Now I'm considering "ESTACIONAMENTO MENSALISTA" and whether to use "MENSALISTA" directly or reuse the existing "MENSALIDADE" stem. Since "MENSALIDADE" doesn't cover "MENSALISTA", I'll shorten the stem to "MENSAL" so it generalizes across mensalidade, mensalista, and mensalmente, checking there's no collision elsewhere. Then I'm weighing whether "PLANO ODONTOLOGICO ODONTOPREV" should fall under Fixos using a "PLANO" stem, comparing it against other plano-related entries like plano saude and plano celular.

For telecom, I'm using stems like VIVO, CLARO, TIM, OI, checking each for substring collisions - CLARO looks safe, but TIM is risky since it's short enough to accidentally match inside words like "ESTIMA" or "OTIMO".

For insurance, SEGURO works cleanly as a stem covering auto, residential, life, and cell phone policies with no collisions elsewhere. For loans, I'm listing stems like FINANCIAMENTO, CONSORCIO, EMPRESTIMO, CONSIGNADO, IPTU, IPVA, and ALUGUEL, and for health plans I'm considering brand fallbacks like UNIMED, AMIL, HAPVIDA, and ODONTOPREV in case the word "PLANO" is missing, though all current examples include it so the stem already covers them.

I'm adding those brand names anyway as extra keywords to improve generalization to descriptions that might omit "PLANO." For gym memberships, ACADEMIA works as a stem covering SmartFit, Bio Ritmo, and Bodytech. For subscriptions I'm working through Netflix, Spotify, Disney, and Amazon Prime, but I need to be careful since a bare "AMAZON" stem would collide with Amazon.com.br retail purchases, so I should use the full "AMAZON PRIME" phrase instead to keep fixed subscriptions distinct from retail spending.

For streaming and software subscriptions, I'm deciding to use full phrases like "YOUTUBE PREMIUM" rather than a generic "YOUTUBE" stem to avoid ambiguity, while also adding GloboPlay, Deezer, Apple's billing identifier (normalized from "APPLE.COM/BILL" to "APPLE COM BILL"), Microsoft 365, and Adobe to the keyword list.

Now I'm adding iCloud, Xbox Game Pass, and PlayStation Plus as specific phrases, confirming these won't collide with other Outros entries like "PLAYSTATION STORE" or "XBOX LIVE" since those don't share the same substrings. Moving on to variable expenses, I'm starting to list supermarket-related stems: SUPERM, HIPERMERCADO, ATACAD, MERCEARIA, MERCADINHO, VAREJA.

Continuing the list with SACOLAO, HORTIFRUTI, QUITANDA, "FEIRA LIVRE", ACOUGUE, PADARIA, PANIFICADORA, POSTO, and pharmacy stems DROG and FARM to cover common drugstore chains, plus convenience store terms like CONVENIENCIA, OXXO, "AM PM", and utility companies SANEPAR and COPEL.

I'm checking each stem for false-positive collisions against other categories — FARM, DROG, POSTO, GAS, AGUA, ENERGIA, and ATACAD all look safe with no overlapping matches elsewhere in the dataset. I'm also deciding to add a broad "SUPER" stem since it's harmless even though it overlaps redundantly with SUPERM matches.

Now I'm organizing the keyword arrays by priority — Investimentos first, then Custos Fixos, then Custos Variáveis — and double-checking that typical "Outros" transactions like IFOOD, UBER, 99POP, and RAPPI entries don't accidentally trigger any of these keyword stems.

Checking more entries: "CHOPERIA CENTRAL" has no match, and "COMPRA MERCADO LIVRE" doesn't trigger since "MERCADO" alone isn't a keyword stem, so it stays Outros. Moving on to check "COMPRA AMAZON.COM.BR" next.

Continuing through more examples: Xbox Game Pass, ticket sites like Ingresso.com and Sympla, travel bookings like 123 Milhas, Decolar.com, Latam and Azul airlines — none trigger keyword matches, so they correctly fall into Outros. I note Azul could also be a fintech name, worth double-checking.

Checking Airbnb, Booking.com, nightclub and show venues, and various food spots like ice cream shops, burger joints, pizzerias, and cafes — none of these match my current keyword list either, so they'd all default to Outros as well.

Checking more edge cases: gift shops, florists, pet shops, salons, barbershops, and tattoo studios don't match any keyword stems, so they correctly default to Outros. I'm also verifying that "BARBEARIA AVULSA" stays Outros while "CLUBE ASSINATURA BARBEARIA" correctly triggers Custos Fixos through the "ASSINATURA" stem rather than the barbershop name itself.

Now I'm going through the Custos Fixos entries to confirm each one matches a fixed-cost keyword before the Investimentos check could intercept it—Netflix, Spotify, Disney Plus, Amazon Prime, and HBO Max all correctly hit their respective subscription keywords without overlapping into investment categories.

Actually simpler to just use "APPLE" as the stem since no other entry collides with it, covering any Apple billing variant. And "MENSAL" works as a stem since it's a prefix of "MENSALIDADE", so it'll match both the Unicesumar and Estacio entries correctly.

Continuing with other categories: CONDOMINIO for condo fees, ALUGUEL for rent, FINANCIAMENTO for financing (covers both real estate and vehicle loans), CONSORCIO for consortium payments, SEGURO for insurance (auto, home, life, phone), and PLANO or the specific provider name for health plans.

I'm also mapping gym memberships to ACADEMIA, telecom carriers like VIVO, CLARO, TIM, and OI to their own brand names, school tuition debits to MENSAL, property and vehicle taxes (IPTU/IPVA) to their own codes, payroll-deducted loans to CONSIGNADO, personal loans to EMPRESTIMO, and software subscriptions like Microsoft 365 and Adobe Creative Cloud to either the brand name or ASSINATURA.

Continuing this pattern, I'm extending the mapping to cover cloud storage (ICLOUD), gaming subscriptions (XBOX GAME PASS, PLAYSTATION PLUS), monthly parking fees and dental plans, subscription clubs, language course tuition, cell phone insurance debits, account maintenance fees, and card annuity charges.

Now I'm verifying the remaining entries—monitored alarm services, TV subscriptions, and controlled cell phone plans—don't accidentally overlap with investment-related keywords like PREVIDENCIA, then moving on to cross-check the Investimentos category entries for brokerage firms like XP and Rico.

Continuing through the list: "BTG PACTUAL DIGITAL" lacks a matching keyword so it needs explicit handling, while "INTER INVEST APLICACAO" and "MODALMAIS CORRETORA" match fine via INVEST/APLICACAO/CORRETORA. I'm checking more examples like TESOURO, CDB, LCI, LCA, and PREVIDENCIA entries to confirm each hits the right keyword pattern. Even when a term like "PORTO SEGURO" could overlap with another category, checking Investimentos before Fixos ensures the correct match wins, which confirms the category order is essential.

Continuing through more examples: FUNDO IMOBILIARIO, FUNDO DE INVESTIMENTO, and APORTE all map to investment-related categories, so I'm adding "APORTE" as a general keyword too. Other patterns like COMPRA ACOES, CORRETORA transfers, APLICACAO POUPANCA, and crypto/investment app transactions (PICPAY INVEST, MERCADO BITCOIN) all reinforce the same keyword groupings.

I'm now checking whether "EASYNVEST" contains the substring "INVEST" -- spelling it out letter by letter to verify, since it's close but not an exact match. It doesn't, but it's still caught by the "APLICACAO" keyword instead, so that's fine. I continue mapping other bank/investment terms like ORAMA, SAFRA, C6, BANCO INTER, and NU INVEST to their respective keyword categories.

More examples: "POUPANCA PROGRAMADA CAIXA" maps to POUPANCA PROGRAMADA, "TRANSFERENCIA TED CORRETORA" maps to CORRETORA, and "APORTE MENSAL PREVIDENCIA" is tricky since it contains both MENSAL and PREVIDENCIA, creating a potential keyword collision I need to watch for.

This confirms that checking Investimentos before Custos Fixos in the cascade order is essential, since "APORTE MENSAL PREVIDENCIA" would otherwise match the wrong category. I'm also mapping "COMPRA ETF BOVA11" to ETF, "APLICACAO RENDA FIXA SOFISTICADA" to APLICACAO, and noting that "BTG PACTUAL DIGITAL" needs an explicit "BTG" keyword added since it wouldn't match otherwise. I should also verify entries like "COMPRA ATACADAO LTDA" for similar issues.

Now I'm scanning the Custos Variaveis examples to make sure none accidentally contain Investimentos keywords like INVEST, APLICACAO, FUNDO, CORRETORA, TESOURO, PREVIDENCIA, CDB, LCI, LCA, ACOES, B3, BITCOIN, CRIPTO, BINANCE, FOXBIT, DEBENTURE, CRA, CRI, ETF, FII, APORTE, or BTG — none do, which is good. Next I need to check Custos Variaveis against Custos Fixos keywords for similar collisions.

None of those health/insurance keywords overlap with Variaveis examples, so that's safe. I'm also verifying the utility keywords like SANEPAR, COPEL, and gas/water/energy terms don't collide with Fixos, Investimentos, or Outros lists, and checking whether "GAS" as a substring might falsely match inside other words across all categories.

I don't find "GAS" embedded elsewhere, and "POSTO" doesn't collide with "ESTACIONAMENTO" since the letters don't line up as a substring. AGUA and ENERGIA also look clean with no accidental overlaps. Now I'm planning test cases covering at least three examples per category plus edge cases to validate the generalization.

I'm also working through more examples for Custos Variaveis and Outros categories, plus edge cases like empty or whitespace-only strings, which should default to "Outros".

I should also test case-insensitivity (uppercase vs lowercase), accented characters being stripped properly, and mixed-case strings with trailing codes. Now I'm ready to start writing the actual code files, beginning with the classification logic.

For testing, I notice there's no test framework installed yet, so I need to add one. Vitest seems like the right choice given it's lightweight, fast, and integrates well with TypeScript/Next.js projects, so I'll justify that addition before proceeding.

The CLAUDE.md header explicitly states these rules override default behavior, so this is a clear, unambiguous instruction I must follow. Since the user asked for tests but didn't specify a library, I'll use AskUserQuestion to quickly confirm which test runner they'd prefer before installing anything.
Antes de tudo: a função de classificação em si não precisa de nenhuma dependência nova (é só string manipulation puro). Mas para os testes automatizados preciso instalar um test runner, já que o projeto ainda não tem nenhum configurado — conforme a regra do CLAUDE.md, vou perguntar antes de adicionar.
PerguntadoTest runner
The user doesn't want to proceed with this tool use. The tool use was rejected (eg. if it was a file edit, the new_string was NOT written to the file). STOP what you are doing and wait for the user to tell you how to proceed.

Note: The user's next message may contain a correction or preference. Pay close attention — if they explain what went wrong or how they'd prefer you to work, consider saving that to memory for future sessions.
questions: [{"question":"O projeto não tem test runner configurado ainda. Qual biblioteca posso adicionar como dependência de desenvolvimento pra rodar os testes automatizados da classifyTransaction?","header":"Test runner","multiSelect":false,"options":[{"label":"Vitest (Recomendado)","description":"Rápido, configuração mínima, integra bem com TypeScript/ESM e é o padrão mais comum em projetos Next.js atuais."},{"label":"Jest","description":"Mais tradicional/consolidado no ecossistema React, porém exige um pouco mais de configuração com TypeScript/ESM."}]}]

Pode usar o Vitest

Comando:

export type Category = | "Investimentos" | "Custos Fixos" | "Custos Variáveis" | "Outros";

/**

- Deixa maiúsculo, remove acentos, troca símbolos por espaço e tira
- números/códigos soltos no final (ex.: "PAG*DROGASIL 0231" -> "PAG DROGASIL"). */ function normalizeDescription(description: string): string { return description .normalize("NFD") .replace(/[\u0300-\u036f]/g, "") .toUpperCase() .replace(/[^A-Z0-9]+/g, " ") .replace(/\s+/g, " ") .trim() .replace(/(?:\s+\d+)+$/, "") .trim(); }

/**

- "Stems" casam como substring solta (ex.: "SUPERM" acha "SUPERMUFFATO"),
- úteis pra generalizar nomes de marcas compostas. */ function matchesAnyStem(normalized: string, stems: string[]): boolean { return stems.some((stem) => normalized.includes(stem)); }

/**

- "Words" só casam como palavra inteira (com espaço/borda dos dois lados),
- pra evitar falso positivo de termos curtos (ex.: "ETF" dentro de "NETFLIX"). */ function matchesAnyWord(normalized: string, words: string[]): boolean { const padded = `${normalized}`; return words.some((word) => padded.includes(`${word}`)); }

const INVESTIMENTOS_STEMS = ["INVEST", "CORRETORA", "APLICACAO", "APORTE", "FUNDO", "TESOURO", "PREVIDENCIA", "DEBENTURE", "BITCOIN", "CRIPTO", "BINANCE", "FOXBIT", "BTG", "POUPANCA PROGRAMADA",]; const INVESTIMENTOS_WORDS = ["CDB", "LCI", "LCA", "ACOES", "B3", "ETF", "FII", "CRA", "CRI"];

const CUSTOS_FIXOS_STEMS = ["ASSINATURA", "MENSAL", "NETFLIX", "SPOTIFY", "DISNEY", "AMAZON PRIME", "YOUTUBE PREMIUM", "GLOBOPLAY", "DEEZER", "APPLE", "MICROSOFT", "ADOBE", "ICLOUD", "XBOX GAME PASS", "PLAYSTATION PLUS", "CONDOMINIO", "ALUGUEL", "FINANCIAMENTO", "CONSORCIO", "SEGURO", "PLANO", "UNIMED", "HAPVIDA", "ODONTOPREV", "ACADEMIA", "CLARO", "CONSIGNADO", "EMPRESTIMO", "ANUIDADE", "MANUTENCAO", "ALARME",]; const CUSTOS_FIXOS_WORDS = ["HBO", "AMIL", "VIVO", "TIM", "OI", "IPTU", "IPVA"];

const CUSTOS_VARIAVEIS_STEMS = ["SUPERM", "HIPERMERCADO", "ATACAD", "MERCEARIA", "MERCADINHO", "VAREJAO", "SACOLAO", "HORTIFRUTI", "QUITANDA", "FEIRA LIVRE", "ACOUGUE", "PADARIA", "PANIFICADORA", "POSTO", "DROG", "FARM", "SUPER", "CONVENIENCIA", "OXXO", "AM PM", "SANEPAR", "COPEL", "COMPANHIA GAS", "AGUA", "ENERGIA", "CARREFOUR", "PAO DE ACUCAR",]; const CUSTOS_VARIAVEIS_WORDS = ["GAS"];

export function classifyTransaction(description: string): Category { const normalized = normalizeDescription(description);

if (!normalized) return "Outros";

if (matchesAnyStem(normalized, INVESTIMENTOS_STEMS) || matchesAnyWord(normalized, INVESTIMENTOS_WORDS)) { return "Investimentos"; }

if (matchesAnyStem(normalized, CUSTOS_FIXOS_STEMS) || matchesAnyWord(normalized, CUSTOS_FIXOS_WORDS)) { return "Custos Fixos"; }

if (matchesAnyStem(normalized, CUSTOS_VARIAVEIS_STEMS) || matchesAnyWord(normalized, CUSTOS_VARIAVEIS_WORDS)) { return "Custos Variáveis"; }

return "Outros"; }

Esse código trata bem uma descrição de transação com caracteres especiais e números soltos, tipo "PAG*POSTOSHELL 1234"? Aponte problemas e sugira melhorias.

Resposta:

*I'm tracing through the normalizeDescription function step by step for "PAG*POSTOSHELL 1234" — normalizing diacritics, uppercasing, replacing special characters with spaces, and stripping trailing numeric suffixes to see what the final cleaned string becomes.*

Then I check it against matchesAnyStem: it ends up matching "POSTO" in CUSTOS_VARIAVEIS_STEMS, which correctly classifies it. But this makes me suspicious of the stem-matching approach generally — substring matching like "POSTO" could produce false positives on unrelated words, and similarly "DROG" as a stem risks matching unintended words beyond "DROGA".

I'm also noticing a potential gap in the trailing-number stripping regex: it only removes digits separated by a space at the end, so if a bank description has no space between the merchant name and a trailing code (like "POSTOSHELL1234" instead of "POSTOSHELL 1234"), the digits won't get stripped and the stem match could fail.

*Also codes embedded mid-string without trailing anchor, like "PAG*POSTO 1234 SHELL", wouldn't get stripped since the regex only anchors at the end. And the fixed cascade order (INVESTIMENTOS → CUSTOS_FIXOS → CUSTOS_VARIAVEIS → OUTROS) means a description matching multiple categories always picks the earliest one, which could misclassify things that legitimately belong elsewhere.*

Short stems also risk false positives — something like "OI" as a standalone word could accidentally match if stripped garbage happens to leave that exact two-letter token, even though word-boundary matching should mostly prevent it from catching substrings like "ADOTO".

I'm spotting a real bug here: multi-word stems like "AMAZON PRIME" or "PAO DE ACUCAR" assume adjacent words with no tokens in between, but real bank descriptions often insert extra words (e.g. "AMAZON PRIME VIDEO BR"), which would break the substring match entirely.

I'll respond in Portuguese, explaining briefly before any code suggestion, per the project convention. Tracing through the example confirms it's correctly classified as "Custos Variáveis" via the "POSTO" stem, so I can confirm this case works while flagging the broader pattern-matching risks. No need to run code for this quick review—I'll write the answer directly.

Rastreando manualmente: `"PAG*POSTOSHELL 1234"` → NFD/uppercase não muda nada → `*` vira espaço → `"PAG POSTOSHELL 1234"` → o regex final remove `" 1234"` → `"PAG POSTOSHELL"`. Isso bate no stem `"POSTO"` (substring) e cai em **Custos Variáveis**. Então esse caso específico funciona, mas o pipeline tem algumas fragilidades que vão te morder em outros extratos:

1. Código sem espaço antes do número não é removido. O regex final é `/(?:\s+\d+)+$/` — exige espaço antes do dígito. Se vier `"PAG*POSTOSHELL1234"` (comum em extratos reais, sem separador), o número gruda no nome e nunca é cortado. Nesse caso específico ainda funciona por acaso (o stem `"POSTO"` é prefixo), mas qualquer `matchesAnyWord` (que exige fronteira de espaço nos dois lados) falha silenciosamente quando o código gruda na última palavra.

2. Números soltos só são removidos no final da string, não no meio. Ex.: `"PAG*1234*POSTOSHELL"` ou `"POSTO 1234 SHELL BR"` mantêm o dígito no meio, o que é inofensivo pros stems atuais mas é uma limitação que vale documentar — o comentário da função promete mais do que o código faz.

3. Stems multi-palavra quebram se houver token extra no meio. `"AMAZON PRIME"`, `"COMPANHIA GAS"`, `"PAO DE ACUCAR"` são checados como substring contígua. `"AMAZON PRIME VIDEO"` funciona (é prefixo), mas `"AMAZON.COM PRIME"` → normaliza pra `"AMAZON COM PRIME"` e o stem `"AMAZON PRIME"` não é mais substring — cai em Outros.

4. Ordem de cascata é rígida e pode mascarar categoria certa. Uma descrição que bate em dois grupos (ex.: contém `"SEGURO"` de Custos Fixos e `"POSTO"` de Custos Variáveis) sempre vai pra Investimentos→Fixos→Variáveis nessa ordem, sem chance de desempate por especificidade.

5. Palavras curtas de 2 letras (`"OI"`, `"GAS"`) são arriscadas. `matchesAnyWord` exige fronteira de espaço, então hoje é seguro, mas qualquer resíduo de sigla de 2 letras isolada por espaço no extrato vira falso positivo — é o tipo de regra que convém revisitar se aparecerem erros de classificação.

Não tenho certeza se extratos reais brasileiros geram exatamente esses padrões (código colado vs. separado por espaço) — isso depende do banco emissor do PDF, então vale você confirmar com uma amostra real antes de eu sugerir uma correção específica pro item 1, que é o mais impactante.

Quer que eu escreva os testes unitários pra esses casos-limite (função já é pura, então é direto testar isolado) antes de mexer na implementação?

07A - arquivo inteiro colado → Custo: US$ 0,23 07B - @arquivo referenciado → Custo: US$ 0,21 Conclusão: a diferença foi pequena porque o @arquivo do Claude Code lê o arquivo inteiro por trás dos panos, igual colar manualmente — a economia de tokens de verdade só aparece quando se referencia um trecho específico em vez do arquivo completo, não apenas ao trocar "colar" por "@arquivo".

Comando 8:

Testei com um extrato real do Banco do Brasil e o parser não reconheceu nenhuma transação (total ficou tudo em "Outros"). O formato real do valor é diferente do que você assumiu: o sinal vem DEPOIS do número, entre parênteses, tipo "498,67 (+)" pra crédito e "20,89 (-)" pra débito — não "-50,00" na frente.

Além disso, cada lançamento tem a descrição (nome de quem recebeu/pagou) numa linha separada da linha com data/lote/documento/valor, geralmente começando com

"Pix - Enviado", "Pix - Recebido" ou "Compra com Cartão" seguido do nome em outra linha.

Segue o texto bruto extraído desse extrato real, cole aqui embaixo:

Extrato de Conta Corrente
Cliente RAFAEL TUDELA RIZENTAL
Agência: 352-2 Conta: 886376-8 Período: 01 a 30/06/2026
Lançamentos
Dia Documento Valor Lote Histórico
29/05/2026 372,79 (+) Saldo Anterior
01/06/2026 498,67 (+) 14397 11149168428702
Pix - Recebido
01/06 11:49 00008863766614 RAFAEL TUDE
01/06/2026 20,89 (-) 13105 60101 Pix - Enviado
31/05 14:37 RAIA DROGASIL SA
01/06/2026 317,48 (-) 13105 60102
Pix - Enviado
31/05 21:32 AUTO POSTO E CONVENIENCIA
01/06/2026 70,00 (-) 13105 60103
Pix - Enviado
01/06 11:49 THOMAS YUDI JARRUS
TANABE
463,09 (+) Saldo do dia
02/06/2026 28,00 (-) 13105 60201
Pix - Enviado

02/06 21:06 LEONEL EUGENIO
BERNARDELI
435,09 (+) Saldo do dia
03/06/2026 70,00 (-) 13105 60301
Pix - Enviado
03/06 08:12 HENRIQUE ZOLIN MEDEIROS
03/06/2026 70,00 (-) 13105 60302
Pix - Enviado
03/06 08:18 THOMAS YUDI JARRUS
TANABE
03/06/2026 132,02 (-) 13105 60303 Pix - Enviado
03/06 08:42 PANVEL FARMACIAS
163,07 (+) Saldo do dia
05/06/2026 50,28 (-) 13105 60501 Pix - Enviado
04/06 09:10 PANVEL FARMACIAS
112,79 (+) Saldo do dia
08/06/2026 2.295,37 (+) 14397 81027582176701
Pix - Recebido
08/06 10:27 08863766614 Rafael Tudela
08/06/2026 560,00 (-) 13105 60801
Pix - Enviado
08/06 10:28 JOSE JACINTO MAIA NETO
08/06/2026 453,16 (-) 13105 60802 Pix - Enviado
08/06 14:42 NUSELECT COFFEES INC
08/06/2026 660,57 (-) 13105 60803
Pix - Enviado
08/06 18:18 UNIVERSIDADE UNICESUMAR -

734,43 (+) Saldo do dia
09/06/2026 74,00 (-) 13105 60901 Pix - Enviado
09/06 15:31 REDE VEGAS
09/06/2026 50,00 (-) 13105 60902
Pix - Enviado
09/06 20:09 INSTITUTO LEGENDARIOS
610,43 (+) Saldo do dia
10/06/2026 9.795,66 (+) 14175 100369078
TED-Crédito em Conta
104 6406 59554385000181 NURA
EMPREENDI
10/06/2026 5.000,00 (-) 13105 61001 Pix - Enviado
10/06 17:56 ANA CLARA TUDELA
10/06/2026 1.843,69 (-) 13158 182994129 Pagto
cartão crédito
ALTUS VISA
3.562,40 (+) Saldo do dia
11/06/2026 10.247,00 (+) 14397 110956497725091 Pix -
Recebido

Extrato de Conta Corrente
Cliente RAFAEL TUDELA RIZENTAL
Agência: 352-2 Conta: 886376-8 Período: 01 a
30/06/2026
Lançamentos
Dia Documento Valor Lote Histórico
11/06 09:56 08863766614 Rafael Tudela

11/06/2026 5.818,29 (-) 13105 61101 Pix - Enviado
11/06 10:00 ANA CLARA TUDELA
11/06/2026 70,00 (-) 13105 61102
Pix - Enviado
11/06 11:08 THOMAS YUDI JARRUS
TANABE
7.921,11 (+) Saldo do dia
12/06/2026 1.338,17 (-) 13105 61201
Pix - Enviado
12/06 11:55 LUIZA RAPOSO NASCIMENTO K
6.582,94 (+) Saldo do dia
15/06/2026 215,00 (-) 99008 152725 Compra com Cartão
14/06 14:38 SILVAN CULT
15/06/2026 116,00 (-) 99008 269433 Compra com Cartão
14/06 19:17 GRUPO MADERO
15/06/2026 140,00 (-) 13105 61501
Pix - Enviado
14/06 12:34 HENRIQUE ZOLIN MEDEIROS
15/06/2026 25,00 (-) 13105 61502
Pix - Enviado
15/06 21:00 THOMAS YUDI JARRUS
TANABE
6.086,94 (+) Saldo do dia
16/06/2026 70,00 (-) 13105 61601
Pix - Enviado
16/06 14:37 THOMAS YUDI JARRUS
TANABE

6.016,94 (+) Saldo do dia
17/06/2026 71,86 (-)   99008 179044 Compra com Cartão
17/06 21:57 POSTO VEGAS
5.945,08 (+) Saldo do dia
18/06/2026 74,00 (-)   99008 151825
Compra com Cartão
18/06 14:23 ESPORTES PERFORMANCE
18/06/2026 70,00 (-)   13105 61801
Pix - Enviado
18/06 08:21 HENRIQUE ZOLIN MEDEIROS
18/06/2026 11,99 (-)   13105 61802
Pix - Enviado
18/06 18:22 Renato Iglesias Bernardin
18/06/2026 60,00 (-)   13105 61803
Pix - Enviado
18/06 21:39 KALEO SPORT TENNIS LTDA
5.729,09 (+) Saldo do dia
22/06/2026 27,48 (-)   13105 62201
Pix - Enviado
20/06 13:06 UBER DO BRASIL TECNOLOGIA
22/06/2026 35,00 (-)   13105 62202
Pix - Enviado
20/06 13:06 THOMAS YUDI JARRUS
TANABE
5.666,61 (+) Saldo do dia
24/06/2026 70,00 (-)   99008 179207 Compra com Cartão
24/06 22:00 BlrFoodLtda

5.596,61 (+) Saldo do dia
26/06/2026 50,00 (+) 14397 262348566095661
Pix - Recebido
26/06 23:48 07568110907 LUCAS AZARIAS

Extrato de Conta Corrente
Cliente RAFAEL TUDELA RIZENTAL
Agência: 352-2 Conta: 886376-8 Período: 01 a 30/06/2026
Lançamentos
Dia Documento Valor Lote Histórico
26/06/2026 50,00 (+) 14397 262348597078112
Pix - Recebido
26/06 23:48 00012314254902 DARTAGNAN G
26/06/2026 50,00 (+) 14397 262349210580192
Pix - Recebido
26/06 23:49 00008278906998 GUSTAVO AZA
26/06/2026 210,00 (-) 99008 143144 Compra com Cartão
26/06 11:59 DaliaFloresE
26/06/2026 26,00 (-) 99008 270919 Compra com Cartão
26/06 19:41 MGA
26/06/2026 140,00 (-) 13105 62601
Pix - Enviado
26/06 16:13 THOMAS YUDI JARRUS
TANABE
5.370,61 (+) Saldo do dia
29/06/2026 364,99 (-) 99008 112594 Compra com Cartão

27/06 03:29 ZIG*FOLKS PUB MARING
29/06/2026 30,00 (-) 99008 249275 Compra com Cartão
29/06 13:41 GRUPO MADERO
29/06/2026 66,89 (-) 99008 365214 Compra com Cartão
29/06 18:06 POSTO AUTOMAR
29/06/2026 280,00 (-) 13105 62901 Pix - Enviado
27/06 15:43 ALDONEI DIAS DOS REIS
29/06/2026 70,00 (-) 13105 62902
Pix - Enviado
27/06 17:55 UNIVERSIDADE UNICESUMAR -
4.558,73 (+) Saldo do dia
30/06/2026 269,50 (-) 99008 180713 Compra com Cartão
30/06 22:25 GRUPO HABANERO
30/06/2026 70,00 (-) 13105 63001
Pix - Enviado
30/06 15:18 THOMAS YUDI JARRUS
TANABE
30/06/2026 4.219,23 (+) S A L D O
Total Aplicações Financeiras
* Saldos por dia Base
Sujeitos a confirmação no momento da contratação
0,00

Ajuste o parseStatementLine.ts pra reconhecer esse formato real, extraindo corretamente descrição e valor (considerando "(+)" como positivo e "(-)" como negativo). Atualize os testes também pra cobrir esse novo formato.

Resposta: 8

> **Thinking** (~3191 tok)
>
> I'm examining the actual record format from the PDF extraction, noting each transaction spans multiple lines with date, value, lot/document numbers on one line, followed by a transaction type label like "Pix - Recebido" and then details on subsequent lines.
>
> Given the short academic timeline, I'm leaning toward a simple line-based state machine rather than an overengineered parser. I need to identify lines to skip entirely (balance summaries, headers, totals) versus date+value lines that mark new transactions, with everything after forming the description until the next transaction or skip line appears.
>

> Given the messiness of real-world PDF extraction, I'm settling on a practical MVP approach: treat each date+value+sign line as a record boundary, grouping everything until the next such marker or a stop keyword like "Saldo do dia" into one transaction. Within each record I'll extract the amount and sign, then match the description against known transaction-type keywords.
>
> I should then filter out lines that are pure noise, like date+time stamps or standalone numeric document codes, keeping any trailing name text on those lines. Testing this against example 2, I trace through a record with a Pix recebido transaction to see how the description "Pix - Recebido ... RAFAEL TUDE" would be extracted after stripping the noise lines.
>
> Since that's a Pix received (income, not expense), it's fine to leave it as "Outros" for category purposes since that's outside this task's scope. Moving to the next record, I see "01/06/2026 20,89 (-)" with "Pix - Enviado" already on the same line, followed by "31/05 14:37 RAIA DROGASIL SA" on the next line.
>
> Stripping the date/time prefix leaves "RAIA DROGASIL SA", so the combined description becomes "Pix - Enviado RAIA DROGASIL SA" which matches the DROG stem and classifies as farmácia under Custos Variáveis. The next

record follows a similar pattern with "AUTO POSTO E CONVENIENCIA", matching the POSTO stem for Custos Variáveis, and then another Pix Enviado record begins.
>
> I also notice a "Saldo do dia" line showing the running daily balance, which isn't a transaction and doesn't start with a date, so my boundary detection based on date prefixes won't catch it as a new record — I need to explicitly filter these out so they don't get appended into the preceding transaction's description by mistake.
>
> I realize lines like "Saldo Anterior" and "S A L D O" should be skipped entirely even though they start with a date, since they're balance markers not transactions. I'm working out a classification scheme for each line - marking them as transaction starts, balance lines, or continuation/description lines based on their patterns.
>
> I realize I don't need the balance line as a terminator since the next date-starting line naturally begins a new record — I just need to exclude its content from the current description. I'm also identifying header noise lines like account headers, column titles, and trailing "0,00" values that should be skipped entirely.
>
> For the parsing logic, when a new transaction start is detected, I finalize the previous record with its

accumulated description and push it to results, then begin a new record. At the end, I finalize the last record, and I'm working out how to strip date/time and numeric-code tokens from content lines before appending them to descriptions.
>
> For line 1 residuals like lote/doc numbers, I should strip pure-digit tokens while keeping meaningful text like "Pix - Enviado" or "Compra com Cartão" intact, since those are real content classifiers rather than numeric codes.
>
> So my general cleaning approach: first remove date/time patterns, then strip any remaining standalone digit runs for lote, doc, and account numbers, then collapse and trim whitespace.
>
> I'm applying this to every content line, concatenating non-empty results while skipping balance/header noise. Testing against examples like "13105 60101 Pix - Enviado" confirms it correctly reduces to "Pix - Enviado", and lines that are purely digits collapse to nothing and get skipped, relying on adjacent lines like "Pix - Recebido" to supply the actual content.
>
> Checking "31/05 14:37 RAIA DROGASIL SA" and "01/06 11:49 00008863766614 RAFAEL TUDE" confirms the date/time and digit-run removal works correctly,

combining with the prior line to form "Pix - Recebido RAFAEL TUDE" as expected. Now I need to make sure balance lines like "463,09 (+)\tSaldo do dia" are explicitly detected and skipped, since they don't start with a date and would otherwise slip through the content-line cleaning logic.
>
> That stray "Saldo do dia" text wouldn't trigger a new transaction since it lacks a leading date, so no double-counting of the amount — but it would leak as junk noise appended onto the previous TANABE transaction's description, corrupting that record's text without affecting the totals.
>
> I should add explicit filtering for these balance/header lines before content-cleaning — matching phrases like "saldo do dia," "saldo anterior," "total aplicações financeiras," "extrato de conta corrente," "agência:," and header rows like "dia documento valor," so they get stripped out entirely rather than polluting adjacent transaction descriptions.
>
> The balance line is already filtered by the "S A L D O" match, so that's fine. But the trailing "0,00" after the disclaimer text would otherwise leak into the prior transaction as content, so I should add a general filter: any

content line that, after trimming, is purely a monetary amount with nothing else, gets treated as noise.
>
> I also need to handle repeated page headers — the "Extrato de Conta Corrente / Cliente / Agência / Lançamentos" block reappears mid-statement between pages, interrupting transaction sequences, so I need to detect and skip those repeated header blocks too.
>
> +) ... Pix - Recebido" actually continues across the page-break header noise, with the line "11/06 09:56 08863766614 Rafael Tudela" belonging to that same transaction as the counterparty name. I need my noise filter to skip header lines like "Extrato de Conta Corrente", "Cliente...", "Agência:...", "Lançamentos" without treating them as record terminators, so the open record stays in progress and the following line appends to it, yielding "Pix - Recebido Rafael Tudela" after cleanup. That means expanding my noise-filter list to include these header phrases.
>
> I'll match these case-insensitively as substrings on the trimmed line, skipping them without disturbing the current record state. There's a small risk that a real counterparty name starting with "Cliente" could false-positive against the "cliente " prefix check, but that's rare enough to accept

for an MVP. Blank lines just get skipped too, and I'm checking that the "Agência:" line also matches correctly.
>
> Adding balance-line and bare-number noise patterns...
>
> So the logic becomes: trim each line, skip blanks without closing the current record, then check if the normalized line matches noise or balance patterns as a substring and skip it entirely if so — this naturally covers the "Saldo Anterior" case too.
>
> Next I handle bare-number-only lines like trailing totals by skipping them, then check the transaction-start regex for date, amount, and sign pattern; if it matches, I finalize the current record by pushing its accumulated description if non-empty, discarding silently otherwise.
>
> Still writing parsing logic...
>
> I realize I need to restructure the parser entirely since records can span multiple lines — rather than parsing line by line, I should rename and rework it into a function that processes the whole text block at once, since the existing empty-description fallback logic won't cleanly carry over.
>
> Writing regex patterns...
>

> Adding noise detection and amount parsing helpers...
>
> Still writing parser logic... I'm also noting a potential issue with reusing a regex that has the global flag set, since `.test()` can behave statefully across calls.
>
> Confirming the non-global regexes are safe since lastIndex only matters for global patterns, so this design checks out. Now I'm tracing through sample transaction records to verify the parsing logic handles tricky cases correctly, starting with a card payment entry.
>
> Since "CARTAO" isn't a recognized keyword, this would likely classify as "Outros" for now, which seems fine since fixing classifyTransaction isn't the current priority. Moving to the TED record, I check the first line's remainder after removing digits—it ends up empty, so nothing gets pushed there.
>
> The second line "TED-Crédito em Conta" has no digits or dates, so it's kept as-is and pushed. The third line strips out the numeric tokens, leaving "NURA" after trimming, and the fourth line "EMPREENDI" gets pushed unchanged, giving a final description of "TED-Crédito em Conta NURA EMPREENDI" with amount +9795.66, which looks correct.
>

> Now I'm moving to the first Pix record and checking how its first line gets cleaned after removing the digit sequences.
>
> The second line "Pix - Recebido" is pushed directly, and the third line has its date/time token "01/06 11:49" stripped first, then the remaining digit sequence removed, leaving "RAFAEL TUDE" after trimming. This gives a final description of "Pix - Recebido RAFAEL TUDE" with amount 498.67, which checks out.
>
> Now I'm verifying the "Saldo Anterior" line is correctly flagged as noise by normalizing it to lowercase and stripping accents.
>
> I'm tracing through the next transaction block: the "Pix - Recebido" header line has its digit tokens stripped to leave just "Pix - Recebido" as the first content part, then the following blank line, "Extrato de Conta Corrente" header, and "Cliente RAFAEL TUDELA RIZENTAL" line are all correctly skipped as noise, and I'm now checking the "Agência" line for similar treatment.
>
> After stripping the account number, I get "Rafael Tudela" as the cleaned description and push it. Then I move to the next line starting with "11/06/2026 5.818,29 (-)" which looks like a new transaction entry to process.

>
> Continuing to parse the remaining description lines, I add "THOMAS YUDI JARRUS" and "TANABE" to the current transaction's description parts. Then the "S A L D O" line appears, which matches the noise pattern, so I skip it entirely and leave the current transaction open.
>
> Since no more real transactions follow, this record only gets finalized at the very end of the loop via pushCurrent(). I need to verify that subsequent lines like "Total Aplicações Financeiras" and "Sujeitos a confirmação no momento da contratação" also match noise patterns so they get skipped via continue rather than appended to the description.
>
> The "0,00" also matches the bare amount pattern and gets skipped, so all trailing noise is filtered out correctly. The record finalizes with description "Pix - Enviado THOMAS YUDI JARRUS TANABE" and amount -70.00, confirming the logic works since pushCurrent() is called after the loop ends.
>
> Now I want to verify the design against another real case - a repeated header appearing mid-document on page 3, between two June 26 transaction blocks, to make sure the parser handles page breaks and header repetition without corrupting the transaction data.

>
> Testing the transaction line regex against "20,89 (-)\t13105 60101 Pix - Enviado" confirms it matches correctly, with the amount and sign parsed properly and the digit tokens "13105" and "60101" stripped from the description, leaving "Pix - Enviado" as the cleaned result.
>
> I'm also checking that the next line continues the description with "RAIA DROGASIL SA" appended, giving a final description of "Pix - Enviado RAIA DROGASIL SA" with amount -20.89, which matches expectations. I verify the regex also handles thousands separators correctly, like "9.795,66" matching as a single amount, and that parseAmount correctly strips periods and converts the comma to a decimal point.
>
> I also confirm "10.247,00" parses correctly to 10247, and that accented characters like "ã" and "é" in descriptions like "Pagto cartão crédito" are preserved fine since the extraction keeps UTF-8 encoding and I'm not uppercasing content that would cause issues.
>
> Next I want to write tests using realistic excerpts from the sample text, then update page.tsx to call parseStatementText(text) instead of the old line-by-line split-and-parse approach. I'll keep the existing raw-line classifiedLines table as-is since it's a separate

supplementary view, and just wire the new transactions useMemo to use parseStatementText directly.
>
> I'm going to rename parseStatementLine.ts to parseStatementText.ts for clarity, creating the new file with the exported parseStatementText function and ParsedTransaction interface, then deleting the old file and its test and updating the import in page.tsx.